

CAPE GIRARDEAU, MO, USA

WMO: 723290

Lat: 37.225N

Lon: 89.571W

Elev: 336

StdP: 14.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	15.4	-0.5	5.4	11.7	4.7	7.1	18.0	26.3	41.5	24.0	43.4	8.2	350	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	93.9	77.1	91.9	76.7	90.0	75.9	80.3	89.9	78.9	88.4	77.7	86.8	8.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	144.6	86.5	76.1	138.1	84.7	74.9	132.7	83.2	44.0	89.5	42.5	88.7	41.2	86.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.4	17.5	DB	0.3	98.1	7.0	2.7	-4.8	100.0	-8.8	101.6	-12.7	103.1	-17.8	105.1	
			WB	-0.7	82.3	6.6	1.5	-5.4	83.4	-9.3	84.3	-13.0	85.2	-17.9	86.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	34.3	38.7	47.5	58.0	67.7	75.9	78.7	77.2	69.7	58.6	46.8	37.9
	DBStd	17.76	11.24	10.89	10.19	8.57	7.52	5.42	4.71	4.91	6.98	8.85	9.20	10.29
	HDD50	1618	501	339	170	26	1	0	0	0	0	21	167	393
	HDD65	4195	951	735	546	238	63	2	0	1	34	238	548	839
	CDD50	4428	15	24	93	267	548	778	889	843	592	289	71	19
	CDD65	1531	0	0	4	29	146	330	424	379	176	41	2	0
	CDH74	15103	0	0	27	214	1204	3340	4415	3780	1705	403	15	0
	CDH80	6041	0	0	1	22	342	1389	1929	1640	624	93	1	0
Wind	WSAvg	7.4	9.0	9.2	9.3	9.4	7.6	6.4	5.3	4.7	5.1	6.6	7.9	8.1
Precipitation	PrecAvg	46.2	3.0	3.2	4.6	4.3	5.2	3.5	3.5	3.6	3.8	3.0	4.2	4.3
	PrecMax	68.3	11.8	9.6	11.9	12.7	16.9	11.1	10.7	7.8	10.2	10.6	8.3	12.3
	PrecMin	29.9	0.5	0.9	0.7	1.3	1.6	0.7	0.6	0.3	0.5	0.0	0.4	0.8
	PrecStd	7.9	2.3	2.0	2.7	2.5	3.3	2.0	2.3	2.0	2.6	2.2	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	70.1	77.5	82.7	90.1	95.7	97.1	96.7	93.1	87.1	76.0	66.8
		MCWB	61.7	63.0	63.7	67.6	73.1	75.6	79.3	77.6	74.7	71.5	63.1	62.3
	2%	DB	62.4	65.0	73.1	79.5	86.8	92.7	93.5	93.5	90.0	82.4	70.4	62.7
		MCWB	58.9	57.9	62.1	66.4	71.8	75.6	78.4	77.3	74.2	67.8	61.5	58.5
	5%	DB	57.2	61.0	68.7	76.3	83.9	90.5	91.5	91.1	87.0	78.7	66.4	58.4
		MCWB	52.9	54.9	59.1	64.3	70.4	74.8	78.0	76.9	73.0	65.5	59.5	54.4
	10%	DB	52.0	56.4	64.5	73.0	80.9	88.1	89.4	88.5	83.6	74.6	62.8	53.9
		MCWB	47.8	50.3	56.5	62.9	69.4	74.1	76.9	75.8	70.8	63.9	56.6	50.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.5	64.2	66.9	71.3	76.3	79.6	82.4	82.0	78.1	73.8	67.2	63.2
		MCDB	64.9	68.9	74.6	79.4	85.3	90.1	92.0	92.5	88.0	83.4	71.9	66.0
	2%	WB	59.4	60.0	63.9	68.8	74.2	78.1	80.8	80.0	76.2	71.1	64.0	59.2
		MCDB	62.0	64.0	71.1	76.1	83.1	88.8	90.0	90.0	86.1	78.9	68.2	62.2
	5%	WB	54.2	56.0	60.8	66.6	72.5	76.8	79.5	78.4	74.6	68.5	60.8	55.1
		MCDB	56.4	59.8	66.8	73.5	81.0	87.0	88.9	87.8	83.4	74.9	65.5	57.7
	10%	WB	47.8	50.6	57.8	64.4	70.9	75.6	78.1	77.2	73.0	65.8	57.1	50.4
		MCDB	51.5	55.1	63.7	71.0	78.7	84.9	86.8	86.0	80.5	72.5	61.8	53.6
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.6	19.8	21.2	19.7	19.9	19.0	21.0	23.5	23.5	19.5	15.8
		MCDBR	20.8	23.1	24.1	23.2	22.4	22.5	20.7	23.5	25.8	26.7	24.3	19.5
		MCWBR	17.5	17.9	15.9	13.0	10.5	8.5	8.4	10.0	11.8	13.7	16.9	16.2
	5% WB	MCDBR	17.6	20.5	21.0	19.2	18.9	19.5	18.7	20.3	21.7	21.4	20.3	17.7
		MCWBR	16.7	17.9	15.6	12.8	10.4	8.8	8.6	9.8	10.6	12.2	16.8	16.3
Clear-Sky Solar Irradiance	taub	0.315	0.328	0.361	0.395	0.431	0.443	0.481	0.476	0.411	0.356	0.322	0.308	
	taud	2.470	2.416	2.384	2.286	2.243	2.223	2.145	2.152	2.319	2.460	2.500	2.494	
	Ebn at Noon	277	287	286	281	271	267	256	254	266	273	272	272	
	Edh at Noon	26	31	36	41	44	45	48	47	37	30	25	24	
All-Sky Solar Radiation	RadAvg	696	889	1223	1602	1807	2051	1976	1807	1514	1140	776	587	
	RadStd	69	113	108	107	151	128	118	98	90	122	84	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131

Nomenclature: See separate page